

Project Description: Product Page Conversion Optimization

This project undertook a comprehensive hypothesis testing approach to optimize product page performance by analyzing the impact of a new layout on two critical metrics: conversion rates and average cart size. The study involved setting up controlled experiments, performing statistical tests, and evaluating effect sizes to derive actionable insights.

A. Analysis Based on Conversions

1. Objective:

The primary objective of this segment of the project was to test the new product page layout to analyze its impact on user conversions, specifically focusing on the 'Add to Cart' conversion rate.

2. Data Loading and Baseline Calculation:

The analysis began by loading relevant data and establishing a baseline. A historic conversion rate was calculated as approximately 0.0949 (or 9.49%), serving as the initial benchmark against which the new layout's performance would be compared.

3. Sample Size Determination (Power Analysis):

To ensure the reliability and statistical power of the A/B test, a minimum required sample size was calculated using NormalIndPower. This analysis determined that a minimum of 50 users per group was needed to detect a statistically significant difference, given a predefined effect size, significance level, and desired power.

4. Generating Control and Treatment Data & Visualization:

For the A/B test, control and treatment group data were generated. Each group consisted of 1000 users, far exceeding the calculated minimum sample size, thus enhancing the test's robustness.

The summary of observed conversion rates from the generated data was:

- **Control Conversion Rate:** 0.095 (9.5%)
- **Treatment Conversion Rate:** 0.196 (19.6%) The standard deviations for control and treatment groups were 0.293 and 0.397 respectively. These conversion rates were also visually represented using a bar plot to highlight the apparent difference.

5. Central Limit Theorem (CLT) Application:

The project conceptually applied the Central Limit Theorem by plotting the sampling distribution of conversion rates. This step is crucial as it underpins the validity of using parametric tests like the Z-test, ensuring that the sampling distribution of proportions approximates a normal distribution given sufficiently large sample sizes.

6. Confidence Interval Calculation:

A 95% Confidence Interval was calculated for the difference in conversion rates between the treatment and control groups.

- The **difference in conversion rates (treatment - control)** was 0.1010.

- The **95% Confidence Interval was (0.0704, 0.1316)**. This interval suggests that we can be 95% confident that the true difference in conversion rates between the new layout and the old one lies within this range, indicating a positive impact.

7. Hypothesis Test (Two-Sample Z-test for Proportions):

A two-sample Z-test for two independent proportions was conducted to formally test the impact of the new layout on conversions.

- **Null Hypothesis (H0):** $CR_{new} = CR_{old}$ (The new layout's conversion rate is equal to the old layout's conversion rate).
- **Alternate Hypothesis (HA):** $CR_{new} > CR_{old}$ (The new layout's conversion rate is greater than the old layout's conversion rate). The test yielded a **Z-statistic of 6.4050** and a **p-value of 0.0000000**. Given the extremely low p-value (which is much less than the typical $\alpha=0.05$), the **null hypothesis was unequivocally rejected**.

8. Effect Size:

To quantify the magnitude of the observed effect, two effect size measures were calculated:

- **Odds Ratio:** 2.32, indicating that the odds of converting with the new layout are 2.32 times higher than with the old layout.
- **Cohen's d:** 0.2894. This value is interpreted as a **small to medium effect**, suggesting a noticeable, but modest, improvement in conversion.

Conclusion (Based on Conversions):

The analysis based on conversions provides strong statistical evidence that the new product page layout led to a statistically significant increase in the 'Add to Cart' conversion rate. The Z-test results (p-value = 0.0000) allowed for the rejection of the null hypothesis, confirming the positive impact. The confidence interval further supports a substantial positive difference, and Cohen's d indicates a small to medium practical significance of this improvement.

B. Analysis Based on Cart Size

1. Objective:

The objective of this part of the project was to test the new product page layout to analyze its impact on the average cart size.

2. Data Loading and Baseline Calculation:

The project established a baseline for cart size, with the historic average cart size recorded as 2.08.

3. Sample Size Determination (Power Analysis):

A power analysis was conducted to determine the required sample size for detecting a desired difference in average cart size.

- A **desired detectable difference of 0.4** was specified.
- The **estimated standard deviation was 1.42**.
- The calculated **Cohen's d (effect size) for this difference was 0.282**. Based on these parameters, the **required minimum sample size per group was determined to be 156**.

4. Generating Control and Treatment Data & Visualization:

Similar to the conversion analysis, control and treatment data were generated for the cart size analysis. The average cart sizes for the generated data were:

- **Control mean cart size:** 2.00
- **Treatment mean cart size:** 2.37 A bar plot was created to visually compare these average cart sizes between the groups.

5. Central Limit Theorem (CLT) Application:

The Central Limit Theorem was applied by plotting the sampling distribution of sample means. This step is critical for validating the use of parametric tests, as it confirms that the distribution of sample means will approximate a normal distribution, regardless of the underlying population distribution,¹ given sufficient sample sizes.

6. Confidence Interval Calculation:

A 95% Confidence Interval was calculated for the difference in mean cart sizes between the treatment and control groups.

- The **difference in means (Treatment - Control) was 0.370**.
- The **95% Confidence Interval was (0.263, 0.477)**. This interval indicates that we can be 95% confident that the true difference in average cart size between the new and old layouts lies between 0.263 and 0.477 units, implying a positive increase.

7. Hypothesis Test (Welch's t-test):

A Welch's t-test (suited for situations with potentially unequal population variances) was conducted to formally test the impact on average cart size.

- **Null Hypothesis (H0):** $\text{size}_{\text{new}} = \text{size}_{\text{old}}$ (The new layout's average cart size is equal to the old layout's average cart size).
- **Alternate Hypothesis (HA):** $\text{size}_{\text{new}} > \text{size}_{\text{old}}$ (The new layout's average cart size is greater than the old layout's average cart size). The test results were:
- **t-statistic = 5.6990**
- **degrees of freedom ≈ 1993.46**
- **p-value (one-tailed) = 0.0000** With an extremely low p-value (much less than $\alpha=0.05$), the **null hypothesis (H0) was rejected**. This means the treatment group (new layout) has a statistically significantly higher cart size than the control group.

8. Effect Size Calculation:

To quantify the practical significance of the difference in cart size:

- **Control mean cart size:** 2.00
- **Treatment mean cart size:** 2.37
- **Cohen's d = 0.2549**. This value is interpreted as a **small effect**.

Conclusion (Based on Cart Size):

The analysis strongly indicates that the new product page layout led to a statistically significant increase in the average cart size. The Welch's t-test (p-value = 0.0000) provides robust evidence to reject the null hypothesis. While statistically significant, the Cohen's d of

0.2549 suggests that this increase, though real, represents a small practical effect.

Overall Project Conclusion:

This project successfully demonstrated that the new product page layout has a statistically significant positive impact on both conversion rates and average cart size. The conversion rate saw a substantial increase (from 9.5% to 19.6%), confirmed by a two-sample Z-test with a small to medium effect size. Similarly, the average cart size increased (from 2.00 to 2.37), which was found to be statistically significant via Welch's t-test, albeit with a small practical effect size. These findings suggest that the new product page layout is a valuable optimization, improving both the likelihood of a conversion and the value of that conversion.